

Syllabus reference

3.6.2 Methods to detect and prevent cyber security threats

Understand and be able to explain the following security measures:

- biometric measures (particularly for mobile devices)
- password systems
- CAPTCHA
- using email confirmations to confirm a user's identity
- automatic software updates.

9.2 Ensuring security

Cyber-security protection

Introduction

You have learned that there are many threats to the security and integrity of data. Threats include social engineering and malware. Cyber security can be used to overcome security threats. In this section you will learn about cyber security.

Cyber security

“Cyber security” is a general term to describe the methods we use to protect computer systems from attacks. There are many types of attack, and so there are many different methods used to ensure cyber security. In this section you will learn about:

- passwords
- biometrics
- identity confirmation
- software updates.

Password systems

Weak passwords can help hackers to break into computer systems. Computer users can improve security by choosing strong passwords and by being aware of social engineering threats.

People who run networks and other secure systems can put measures in place to improve password systems. For example, they can require users to use strong passwords. They can make users change their passwords regularly.

Biometric measures

A biometric check is an alternative to a password. Biometric means a record of a physical characteristic. It is used to confirm identity.

Checking physical features needs special equipment and software. For example:

- a thumb or palm print reader
- an iris scanner
- facial recognition software.

A biometric check makes it difficult for criminals to pretend they are someone else. You can steal a mobile device and find out someone's password, but you cannot copy the person's physical features. For this reason, biometrics are often used nowadays to protect mobile devices such as smartphones and tablets. If a device like this is stolen, the criminal cannot use it.



↑ A palm scan proves a person's identity

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CAPTCHA

CAPTCHA is a system that checks whether a user is a real person. When you log on to a website you set up an identity and password. You may also need to complete a CAPTCHA check.

Automated systems find it difficult to read distorted text. CAPTCHA usually includes distorted text. You type in the text that you see. The idea is that only a human reader can read the text.

Email confirmation

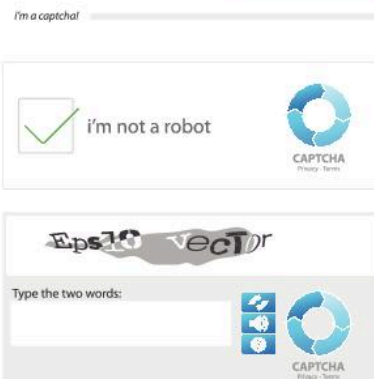
Another way to confirm identity is by email. When you first log on to a system you set up an identity. You usually have to provide your email address. The site will send an email to that address. This is to check that your identity matches the address you have given.

Typically, you will receive an email at that address. It is sent by automated software. It usually comes quite quickly. The email will include a link. You must click on the link to confirm your identity.

Automatic software updates

Anti-malware software must be kept up to date. New viruses are being developed all the time. Anti-virus software must be updated with information about the new viruses that have been developed.

Cyber security depends on up-to-date software. Automatic updates are sent through your Internet connection. The updates come from the company that you bought the software from. An update will bring your software up to date, so that you have better cyber security.



Test yourself

1. I bought anti-virus software two years ago. Why do I need to update it?
2. What is CAPTCHA? How does the use of CAPTCHA help to ensure the security of websites?
3. What are the advantages and disadvantages of biometric methods compared to the use of passwords for security?
4. A college technician wanted to make the password system as safe as possible for the students. What automatic rules might she put in place to ensure good password practice?

Learning activity

Write a report on biometric methods used with modern mobile devices. Look for examples on the Internet.

Extension activity

A school wanted to set up a security system at the entrance to the college. The three methods being considered were that visitors would have to:

- scan an ID card at the gate
- type a passcode on a keypad
- use a palm print reader.

Write a report to the college setting out the advantages and disadvantages of each of these methods.

Penetration testing

Introduction

You have learned about a range of methods used to ensure cyber security. It is important that cyber security measures are kept up to date. They must work perfectly. For this reason, cyber security measures are subject to regular testing. This is called penetration testing.

Internal and external threats

Threats to the security of a system can be internal or external threats:

- **External threats** are attacks on a computer system by hackers outside the organisation. The hackers do not have authentication or passwords. They will try to break the security system. Their attacks come from outside the firewall.
- **Internal threats** occur when people who have bad intentions can get inside an organisation's security system. They might get a job working for the company. They might become trusted. They may have their own password and read/write access.

Internal threats can be much more dangerous and harmful than external threats. Internal threats can be harder to spot. Good hiring practices will help to prevent the wrong people getting access to the system. Good supervision and management will help to prevent this type of attack. Cyber security must protect against internal and external threats.

What is penetration testing?

An attack on system security is an attempt to get access to the data stored on the system. Sometimes hackers want a copy of the data. Sometimes they want to delete or change data. Cyber-security methods try to prevent this.

Penetration testing is a way of testing cyber security. The people who are doing the testing try to break into the system. However, they are not hackers. They are computer experts who work for the organisation. They try to break into the system to see whether it is possible. If the testers can manage to break in, this shows there is a problem with cyber security. The testing has found the problem. Now the computer experts inside the organisation can fix the problem.

A penetration test is sometimes called a pen test for short. There are two main types of pen test. They are called black box and white box testing.

Black box and white box testing

- **Black-box testing** is testing for external threats. The testers work from outside the firewall. They do not use valid IDs or passwords. They break in using an external Internet connection.
- **White-box testing** is testing for internal threats. The testers have some inside knowledge – for example they might have a valid username and password. They might know the structure of the local network. They will try to cause damage to the computer system from inside the organisation. Of course, they do not do real damage.

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Ordinary password testing will not be enough for white-box testing. An internal threat comes from a person who has a password and ID. Cyber-security methods against internal threats would include looking for suspicious patterns of behaviour. For example, an attempt to extend access rights.

Extension – black, white and grey hats

You have learned that the term “hackers” is generally used to describe people who try to break into computer systems. The term is sometimes used to mean a person who likes to investigate and work with computer systems to understand them better. To make things clearer people sometimes divide hackers into different categories:

- Black hat hackers are hackers who break into computer systems. If they find a problem with the security system they will exploit it. They are breaking the law.
- White hat hackers are people who have permission to work with computer systems. They are carrying out penetration testing. They will let the owners of the system know they are testing it. If they find a problem they will report it so it can be fixed.
- Grey hat hackers do not fit into the two previous categories. They may work without permission. However, they may not do any damage. They may let the owners know if they find a problem. Grey hat hackers are breaking the law. However, the main priority of the police will probably be stopping the hackers who cause damage.

Some hackers cause a lot of damage to computer systems. In extreme cases this can lead to financial loss or even harm to people. Although it is not publicised, it may be that some governments use hackers to spy on other governments’ computer systems.

Test yourself

1. How does penetration testing help to improve the security systems of an organisation?
2. Which type of testing will include trying to get past the firewall?
3. What methods might you use to detect a cyber-security threat from inside an organisation?
4. Someone tried to hack into a computer network, but she wasn’t breaking the law. Explain how this could be the case.

Learning activity

Imagine you were given the job of testing the security of your school or college system. What methods might you use to test the security? Write a short plan of action.

Extension activity

Write a Python program that checks a password to see whether it is weak or strong. For greater challenge, display a password report that analyses the password using several criteria.